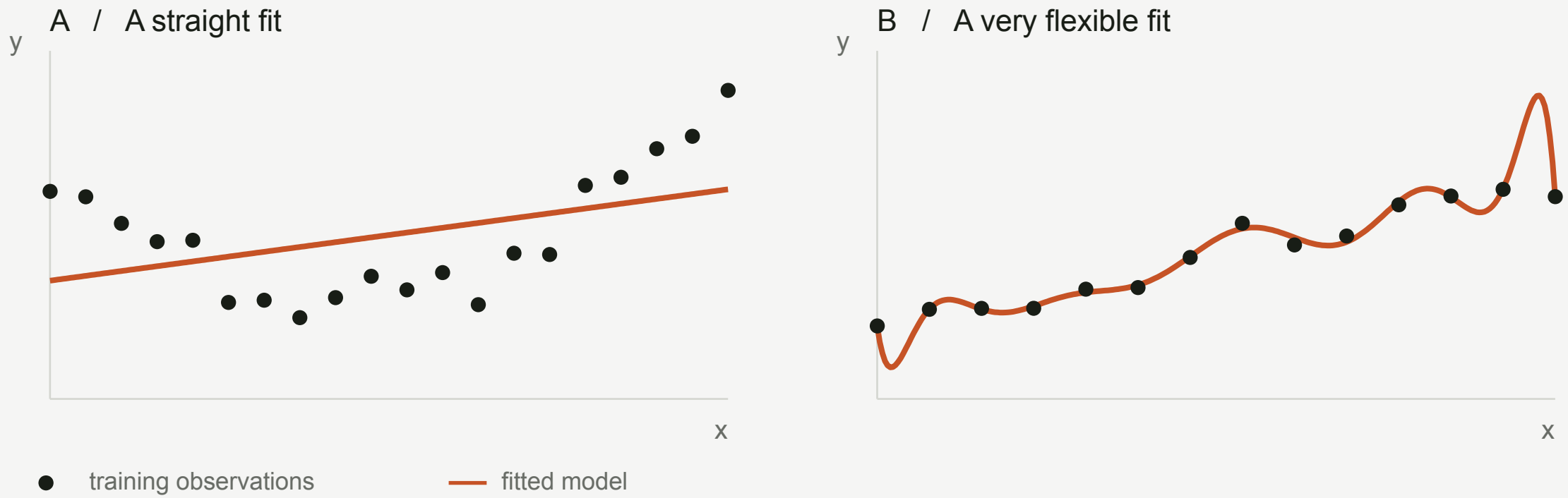
START WITH A PREDICTION

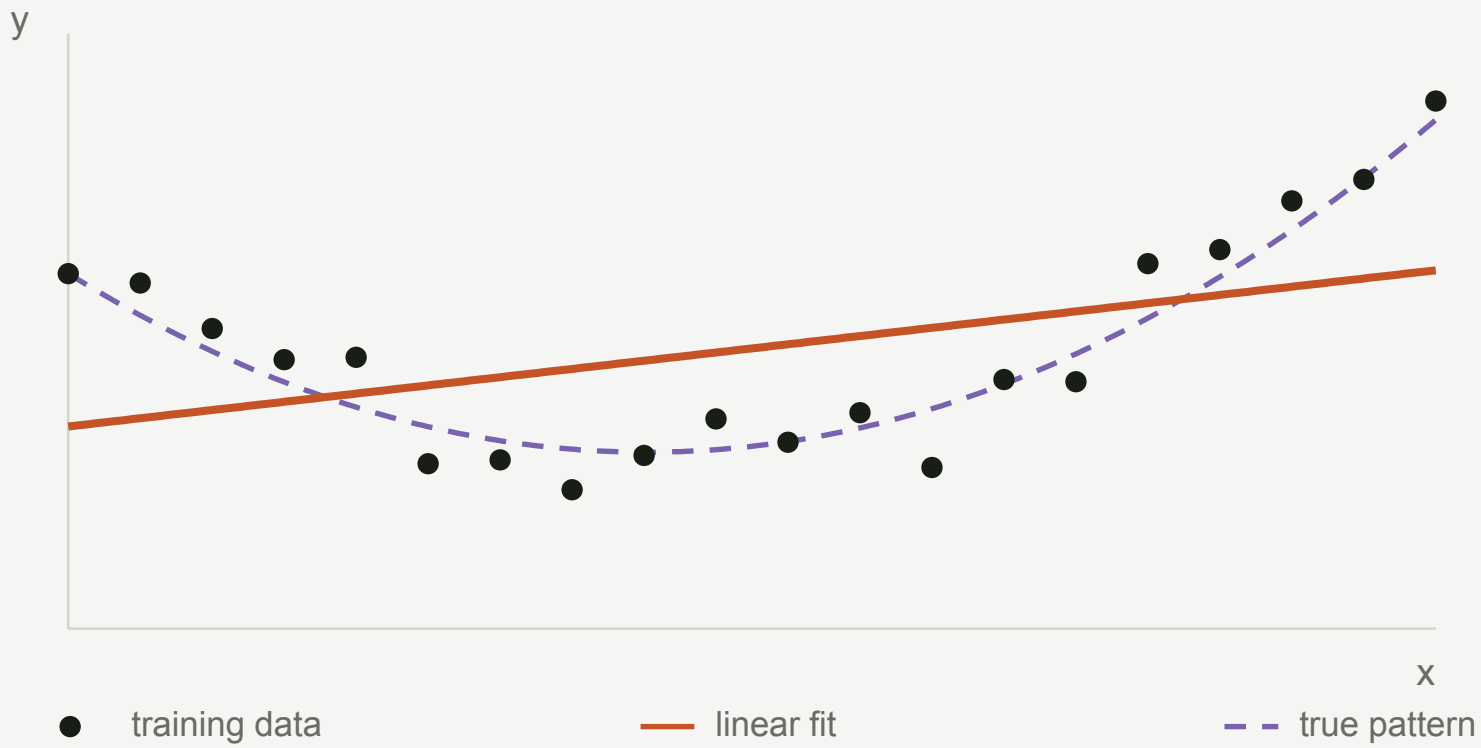
Which model would you trust?

Two different datasets. Two ways a fit can go wrong.



EXAMPLE A / UNDERFITTING

Too simple to learn the pattern.

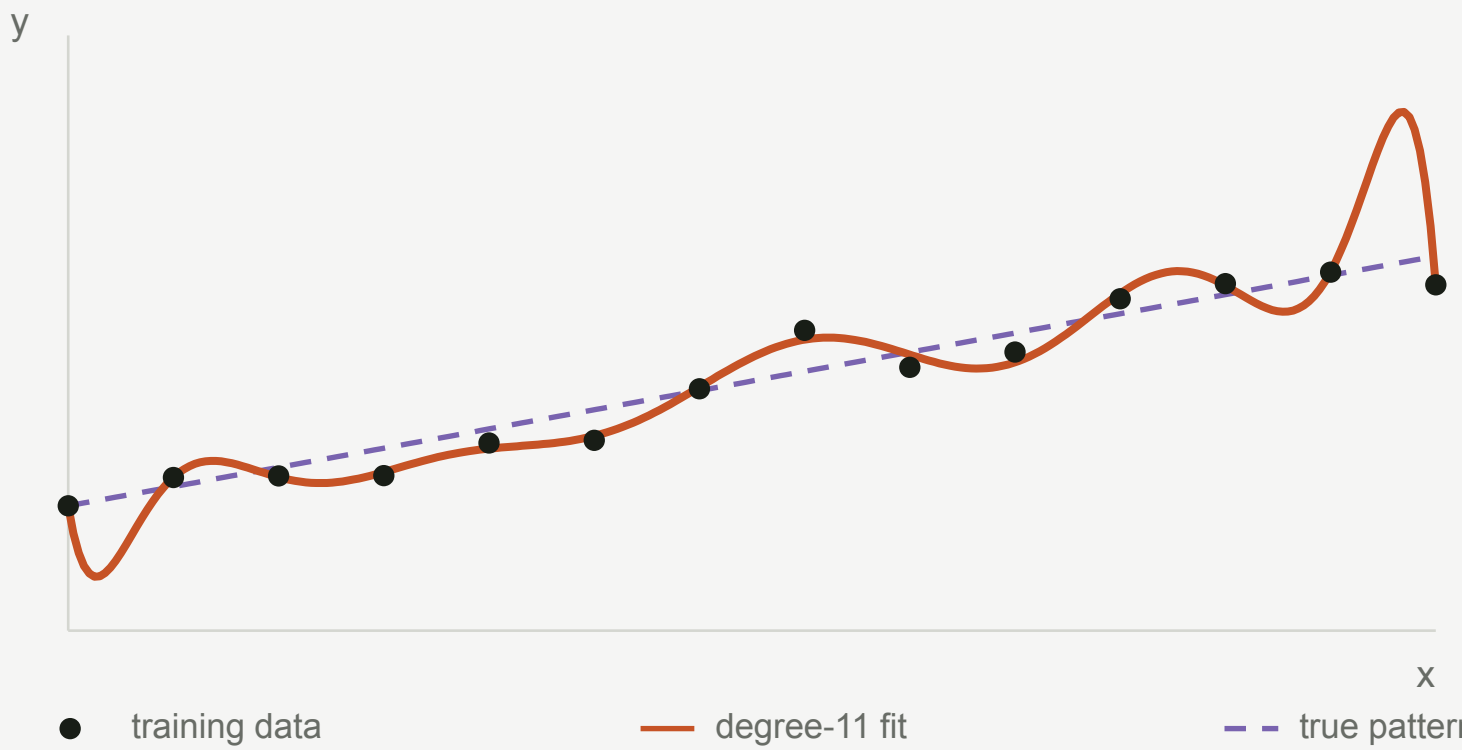


High bias

A straight line cannot learn this curve.

EXAMPLE B / OVERFITTING

Flexible enough to learn the noise.



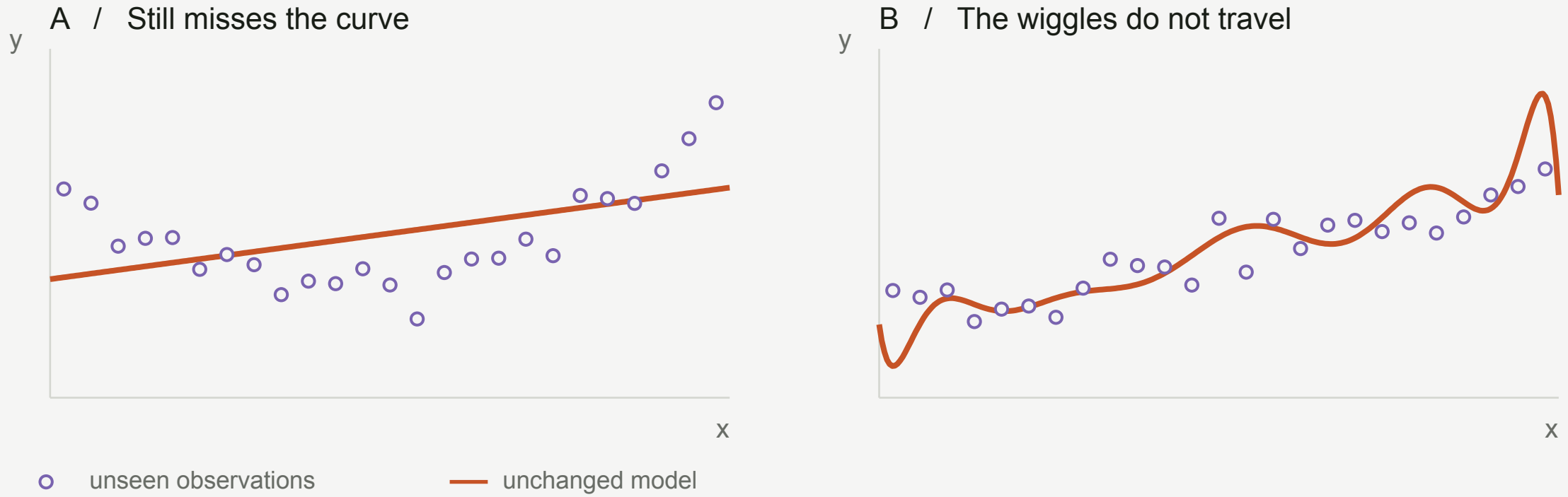
High variance

Small changes in data.
Big changes in the fit.

THE REAL TEST / UNSEEN OBSERVATIONS

New data does not repeat the noise.

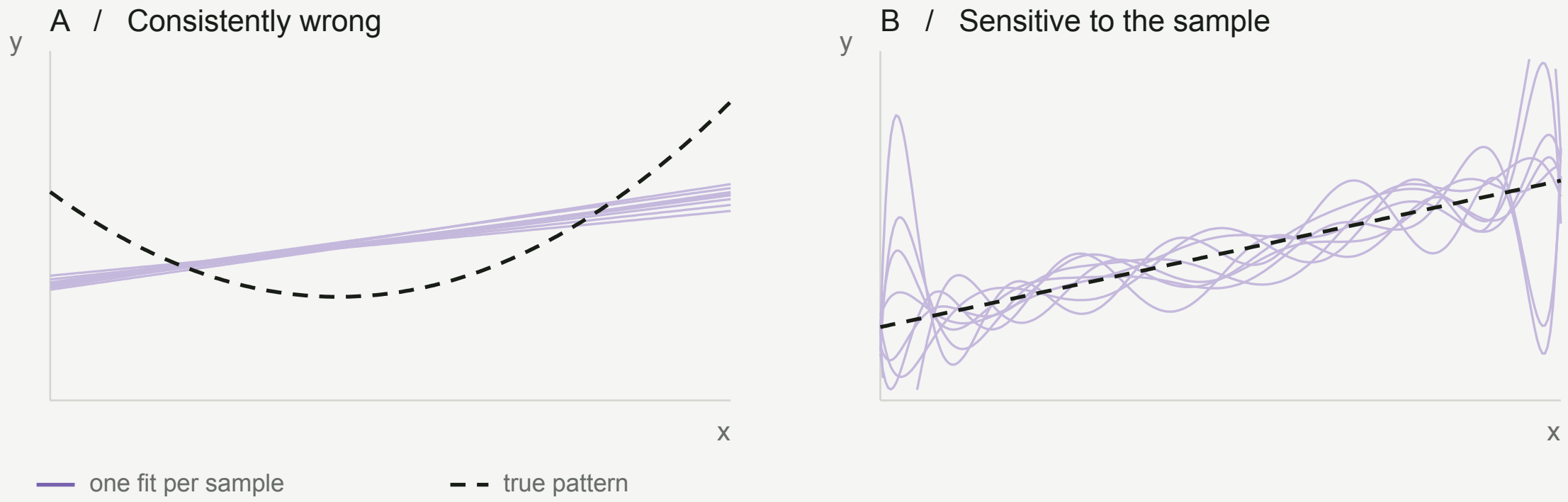
Keep both models frozen. Reveal fresh observations.



WHAT VARIANCE ACTUALLY MEANS

Change the sample. Fit again.

Eight fresh training samples per problem. Same model settings.



CHOOSING COMPLEXITY

A closer fit is not always a better model.



Choose using validation data.

Schematic, not measured

READ THE LOSSES

The smallest training loss can lose.

One curved dataset. Three models. Mean squared error ↓

Fit	Model degree	Training loss	Testing loss
Underfitting	1	0.251	0.183
Good fit	**2**	**0.021**	**0.027**
Overfitting	12	0.008	0.043

Better on training. Worse on testing.

20 training points · 1,000 independent test points · identical split for all models

TAKE IT WITH YOU

Learn the pattern. Not the noise.

BIAS

Too rigid to capture the relationship.

VARIANCE

Too sensitive to the training sample.

GENERALIZATION

How well it predicts new observations.

Tune on validation. Report on the untouched test set.